

VAE Latent Space Visualizer - User Guide

Overview

This interactive tool helps you understand how Variational Autoencoders (VAEs) learn structured latent representations through KL divergence regularization. It visualizes the key concepts from Lecture 24 on VAEs.

Step-by-Step Tutorial

Part 1: Understanding the Baseline (Standard VAE)

Initial Setup:

1. Open the visualizer
2. Ensure all sliders are at their default positions:
 - Mean $\mu_1 = 0$
 - Mean $\mu_2 = 0$
 - Log(σ_1) = 0
 - Log(σ_2) = 0
 - KL Weight $\beta = 1.0$

What you should see:

- Blue dots (samples from $q_\phi(z|x)$) overlapping with gray dots (prior $p(z) = N(0, I)$)
- KL Divergence ≈ 0
- Both distributions centered at origin with unit variance

Key observation: When the encoder outputs $\mu = 0$ and $\sigma = 1$, the learned distribution perfectly matches the prior!

Part 2: The Traditional Autoencoder Problem (Lecture Page 2)

Experiment Setup:

1. Set **KL Weight $\beta = 0.0$**
 - Notice the warning: "Traditional Autoencoder Mode"

- Observe that "Weighted ($\beta \times D_{KL}$)" becomes 0

2. Now move the **Mean μ_1** slider to +2

3. Move the **Mean μ_2** slider to -1.5

4. Adjust **Log(σ_1)** to 0.8

What you should observe:

- The blue dots move to completely different regions of the latent space
- Raw D_{KL} is large (showing deviation from prior)
- BUT: Effective penalty = 0
- The blue distribution looks nothing like the gray prior

Critical insight: Without KL regularization, the encoder can map inputs to arbitrary locations. There's "no restriction" on the latent space (as mentioned in Lecture Page 2).

Questions to consider:

- Why is this problematic for generation? (Hint: what if you sample from $p(z) = N(0, I)$ but the encoder never maps anything there?)
 - Why would interpolation between points be unreliable?
 - If different inputs map to $\mu = [100, 50]$ and $\mu = [-30, 80]$, what does the path between them represent?
-

Part 3: KL Divergence as Regularization

Experiment Setup:

1. Keep the parameters from Part 2 ($\mu_1 = 2$, $\mu_2 = -1.5$, $\text{Log}(\sigma_1) = 0.8$)
2. Gradually increase **β from 0 $\rightarrow$ 0.5 $\rightarrow$ 1.0 $\rightarrow$ 1.5**

What you should observe:

- As β increases, the effective penalty increases
- This represents increasing pressure during training to move parameters back toward $N(0, I)$

Question: Imagine you're training a VAE and the encoder currently outputs these parameters. For each β value:

- What gradient signal does the encoder receive?
- Would the encoder adjust its parameters? In which direction?

- What's the trade-off being made?
-

Part 4: Understanding the Mean Component

Experiment Setup:

1. Reset all parameters to 0
2. Set $\beta = 1.0$
3. Slowly move μ_1 from $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$

What you should observe:

- Blue distribution shifts away from gray prior
- KL divergence increases (watch the raw D_{KL} value)
- The penalty term μ_1^2 grows quadratically

Examine the formula:

$$D_{KL} = \frac{1}{2} \sum [\mu_j^2 + \sigma_j^2 - 1 - \log(\sigma_j^2)]$$

Questions:

- Why does the KL penalty include μ^2 ? What does this term penalize?
 - Why is the penalty quadratic rather than linear in μ ?
 - At what value of μ does the penalty become "large"? Does this make sense for a unit Gaussian prior?
-

Part 5: Understanding the Variance Component

Experiment Setup:

1. Reset to $\mu_1 = 0, \mu_2 = 0, \beta = 1.0$
2. Move $\text{Log}(\sigma_1)$ from $0 \rightarrow 1 \rightarrow 2$
3. Reset, then move $\text{Log}(\sigma_1)$ from $0 \rightarrow -1 \rightarrow -2$

What you should observe:

When $\text{Log}(\sigma) > 0$ ($\sigma > 1$):

- Blue dots spread out (larger variance)
- KL divergence increases
- The term σ^2 dominates the penalty

When $\text{Log}(\sigma) < 0$ ($\sigma < 1$):

- Blue dots cluster tightly (smaller variance)
- KL divergence also increases!
- The term $-\log(\sigma^2)$ dominates the penalty

Critical insight: The KL term penalizes BOTH too much variance AND too little variance!

Questions:

- Why does the VAE penalize $\sigma \neq 1$ in both directions?
 - What would happen if the encoder could set $\sigma \rightarrow 0$? (Hint: this would make z deterministic)
 - Why is it important that z remains stochastic during training?
 - Look at the formula: which term prevents $\sigma \rightarrow \infty$? Which prevents $\sigma \rightarrow 0$?
-

Part 6: The Reparameterization Trick

Background: The visualizer implements: $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}$ where $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \mathbf{I})$

Experiment Setup:

1. Set $\mu_1 = 1$, $\mu_2 = -0.5$, $\text{Log}(\sigma_1) = 0.5$, $\text{Log}(\sigma_2) = 0.3$
2. Click "Resample Points" multiple times

What you should observe:

- Different random samples, but same distribution shape
- The distribution is centered at $(\mu_1, \mu_2) = (1, -0.5)$
- The spread is determined by σ

Questions:

- Why do we need the reparameterization trick? (Hint: can you backpropagate through a random sample?)
- The noise $\boldsymbol{\varepsilon}$ is sampled fresh each time, but $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are learned parameters. Why is this distinction important?

- How does this trick allow us to compute gradients $\partial \text{ELBO} / \partial \mu$ and $\partial \text{ELBO} / \partial \sigma$?
-

Part 7: β -VAE Trade-offs

Experiment Setup: Compare three scenarios with $\mu_1 = 1.5$, $\mu_2 = 0.5$, $\text{Log}(\sigma_1) = 0.6$, $\text{Log}(\sigma_2) = 0.4$:

1. $\beta = 0.0$ (Traditional AE)
2. $\beta = 1.0$ (Standard VAE)
3. $\beta = 2.0$ (β -VAE)

What you should observe:

- Same raw D_KL across all three
- Very different effective penalties
- Different "pressures" on the encoder

Questions:

- With $\beta = 0$, would the encoder ever adjust these parameters? Why or why not?
 - With $\beta = 2$, the penalty is doubled. What effect would this have on:
 - Reconstruction quality?
 - Generation quality?
 - Latent space structure?
 - Why might researchers use $\beta > 1$ in practice? (Hint: disentanglement)
-

Part 8: Connecting to the Lecture (Matrix Formulation)

Refer to Lecture Source 76: The general KL formula is:

$$D_{\text{KL}} = \frac{1}{2} [\text{Tr}(\Sigma) + \mu^T \mu - k - \log \det(\Sigma)]$$

In our 2D case with diagonal Σ :

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

Questions:

- Calculate $\text{Tr}(\Sigma)$ using the current slider values. Does it match $\sigma_1^2 + \sigma_2^2$?
- Calculate $\mu^T \mu$. Does it match $\mu_1^2 + \mu_2^2$?
- For diagonal Σ , what is $\det(\Sigma)$? Does $\log \det(\Sigma) = \log(\sigma_1^2) + \log(\sigma_2^2)$?
- Verify the formula simplification by hand with specific slider values

Exercise: Set $\mu_1 = 1$, $\mu_2 = 0.5$, $\sigma_1 = 1.5$, $\sigma_2 = 0.8$ and calculate:

1. $\text{Tr}(\Sigma) = ?$
 2. $\mu^T \mu = ?$
 3. $\log \det(\Sigma) = ?$
 4. $D_{\text{KL}} = ?$
 5. Compare with the displayed value
-

Discussion Questions for Study Group

Conceptual Understanding

1. Latent Space Structure:

- Why do we want the latent space to look like $N(0, I)$?
- What goes wrong in generation if $q_\phi(z|x)$ and $p(z)$ don't match?
- How does KL regularization enable meaningful interpolation?

2. The ELBO Objective:

- What is the encoder trying to maximize? The decoder?
- Why is there a tension between reconstruction and regularization?
- What happens to the ELBO when $\beta = 0$? Is it still a valid ELBO?

3. Training Dynamics:

- If the encoder outputs $\mu = [5, 3]$ and $\beta = 1$, what gradient signal does it receive?
- Why might KL collapse happen ($D_{\text{KL}} \rightarrow 0$ with $\sigma \rightarrow 0$)?
- How does the reconstruction term prevent complete collapse to the prior?

Mathematical Deep Dive

4. Diagonal Covariance Assumption:

- What would a full covariance matrix allow that diagonal doesn't?
- Why is diagonal Σ computationally attractive?
- What geometric constraint does diagonal Σ impose on the latent distribution?

5. KL Divergence Properties:

- Is D_{KL} symmetric? (i.e., does $D_{KL}(p||q) = D_{KL}(q||p)$?)
- Why do we use $D_{KL}(q||p)$ instead of $D_{KL}(p||q)$?
- What is the minimum value of D_{KL} ? When is it achieved?

6. Reparameterization:

- Write out the gradient $\partial z / \partial \mu$. Is it well-defined?
- Write out the gradient $\partial z / \partial \sigma$. Is it well-defined?
- Could we compute these gradients without the reparameterization trick?

Practical Considerations

7. Hyperparameter Selection:

- How would you choose β for a new dataset?
- What symptoms would indicate β is too low? Too high?
- How does latent dimensionality k affect the appropriate β value?

8. Comparison with Alternatives:

- How does a VAE differ from a GAN in terms of latent space structure?
- What advantages does the VAE's probabilistic formulation provide?
- When might you prefer a traditional autoencoder over a VAE?

Challenge Problems

Challenge 1: Analytical Verification

Pick specific values for all parameters and verify the KL calculation by hand:

- Choose: $\mu_1 = 1$, $\mu_2 = -0.5$, $\sigma_1 = 1.2$, $\sigma_2 = 0.8$
- Calculate D_{KL} using both formulas (trace/determinant and summation)

- Verify your answer matches the visualizer

Challenge 2: Gradient Descent Simulation

Starting from $\mu_1 = 2$, $\mu_2 = 1.5$, $\sigma_1 = 1.5$, $\sigma_2 = 1.5$, $\beta = 1$:

- Estimate the gradient $\partial(\beta \times D_{\text{KL}})/\partial\mu_1$
- If we take a gradient step with learning rate 0.1, where would μ_1 move?
- Simulate several steps manually and check if you approach the prior

Challenge 3: Optimal Parameters

For a given β value:

- What values of μ and σ minimize the effective KL penalty?
 - Is this a local or global minimum?
 - What does this tell you about where the encoder "wants" to map data?
-

Key Takeaways

1. **Without KL regularization ($\beta=0$)**, the latent space is arbitrary and unsuitable for generation
 2. **KL divergence measures** how different $q_{\phi}(z|x)$ is from the prior $p(z) = N(0, I)$
 3. **The reparameterization trick** ($z = \mu + \sigma \odot \epsilon$) enables backpropagation through stochastic sampling
 4. **The β parameter** controls the trade-off between reconstruction quality and latent space structure
 5. **Both mean and variance** are penalized: μ^2 pushes mean toward 0, and $(\sigma^2 - 1 - \log \sigma^2)$ pushes variance toward 1
 6. **Diagonal covariance** simplifies the KL formula from matrix operations to a sum over dimensions
-

Additional Resources

- Review Lecture 24, especially pages 2 (Traditional vs VAE perspective) and page with Source 76 (KL formula derivation)
- Try implementing the Box-Muller transform yourself to understand Gaussian sampling
- Experiment with different β values to build intuition about the reconstruction-regularization trade-off

- Consider how this visualization would extend to higher-dimensional latent spaces ($z \in \mathbb{R}^k$ for $k > 2$)
-

For questions or discussion, share this visualizer with your study group and work through the experiments together!